MUTUAL FUND INVESTMENT & PERFORMANCE ANALYSIS (2019–2024) USING PYTHON

Project Overview

This project analyzes AMFI mutual fund data from 2019 to 2024 using Python to understand mutual fund investment and performance patterns.

The analysis focuses on:

- Mutual fund scheme categories
- Investor participation
- Fund mobilization
- Repurchase and redemption
- Net inflow and outflow
- Net Assets Under Management (Net AUM)
- Average Net AUM
- Changes across reporting years
- Differences between scheme types and categories

The project combines Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis, and Data Visualization to transform raw mutual fund data into meaningful business insights.

Objectives

The main objective of this project is to analyze AMFI mutual fund data using Python and identify useful patterns in mutual fund investment and investor participation.

Key Objectives

1. Identify which mutual fund categories attract more investors and investments.
2. Analyze how **Net AUM changes over time**.
3. Compare mutual fund performance across different scheme types and categories.
4. Identify categories with higher **fund mobilization and redemption**.
5. Analyze positive and negative **net inflow/outflow** patterns.
6. Examine investor participation using the **number of folios**.
7. Study the relationship between **number of folios and Net AUM**.
8. Analyze investment activity across different reporting periods.
9. Identify categories that consistently attract investments.
10. Generate data-driven business insights through statistical analysis and visualization.

Problem Statement

The raw mutual fund dataset does not directly show which scheme categories perform better, how investment activity changes over time, or where positive and negative fund flows occur.

Therefore, this project analyzes AMFI mutual fund data using Python to identify:

- Mutual fund performance
- Investment trends
- Investor participation
- Fund-flow patterns

- Differences across scheme categories
- Changes across reporting periods

The analysis uses **statistical techniques and visualizations** to convert raw data into meaningful business insights.

Business Understanding

The analysis is designed to answer the following business questions.

Fund Performance

- Which scheme category has the highest Net AUM?
- Which scheme type performs better?
- How does AUM change over time?

Investment Activity

- Which categories receive the highest fund mobilization?
- Which categories have the highest redemption?
- Which categories have positive or negative net inflow/outflow?

Investor Participation

- Which categories have the highest number of folios?
- How does investor participation change over time?
- Is there a relationship between folios and AUM?

Trend Analysis

- How does mutual fund activity change across reporting periods?
- Are there periods with unusually high or low investment activity?
- Which categories consistently attract investments?

Dataset Information

Attribute	Details
Data Source	India's Extensive Public Datasets
Location	India
Domain	Economy
Period	2019–2024
Records	3,115
Attributes	14
File Format	Excel
Environment	Google Colab

Attribute / Data Dictionary

Column	Description
Id	Unique identifier for each record
Date	Reporting date of the mutual fund data
scheme_name	Name of the mutual fund scheme
scheme_type	Type of scheme such as Open Ended, Close Ended, or Interval Schemes
scheme_category	Category of the mutual fund scheme
no_of_schemes	Number of schemes
no_of_folios	Number of investor folios
fund_mobilized	Amount of funds mobilized
repurchase_redemption	Amount involved in repurchase/redemption
net_inflow_outflow	Net movement of funds
net_aum	Net Assets Under Management
avg_net_aum	Average Net Assets Under Management
no_of_segregated_portfolios	Number of segregated portfolios
net_aum_in_segregated_portfolios	Net AUM held in segregated portfolios

The dataset contains **56 scheme names, 3 scheme types, and 5 scheme categories**.

Tools & Technologies

Programming Language

- Python 3.10+

Libraries

- **Pandas** – Data manipulation and analysis
- **NumPy** – Numerical operations
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization

Development Environment

- **Google Colab**

Project Workflow

Data Collection

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Initial EDA

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Data Inspection

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Data Cleaning

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Data Validation

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Feature Engineering

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Final EDA

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Statistical Analysis

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Univariate Analysis

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Bivariate Analysis

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Multivariate Analysis

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Business Insights

Data Pre-Processing

The dataset was systematically inspected and cleaned before performing statistical and visual analysis.

Initial Data Inspection

The following checks were performed:

- Dataset shape
- First and last records
- Sample records
- Data types
- Missing values
- Duplicate IDs
- Unique values
- Value counts

- Descriptive statistics
- Special characters
- Zero values
- Negative values

Dataset Shape

Rows : 3,115

Columns : 14

ID Cleaning

The id column was checked for:

- Data type
- Missing values
- Duplicate values

No missing IDs or duplicate IDs were found. The column was subsequently converted from integer to string.

Scheme Name Cleaning

Special characters such as: #,$

were identified in some scheme names and removed to maintain consistent categorical values.

Zero-Value Handling

Zero values were not automatically treated as missing values because a zero can represent a genuine observation in the dataset.

Data Validation

Net Inflow/Outflow Validation

The following relationship was verified:

Net Inflow/Outflow = Fund Mobilized - Repurchase/Redemption

A temporary calculated_net_flow feature was created to validate the existing net_inflow_outflow column.

The maximum absolute difference was approximately: 0.010000000043874024

This represents a very small floating-point difference, confirming that the relationship is consistent.

The temporary validation column was removed after verification.

Feature Engineering

Feature engineering was used to derive meaningful variables for classifying fund flows and investment activity and for supporting year-wise analysis.

Derived Feature	Purpose	Result
Net Flow Status	Classify whether net inflow/outflow is positive, negative, or neutral.	Positive: 1,633; Negative: 1,157; No Flow: 325
Investment Status	Compare fund mobilization with redemption to identify investment behavior.	More Mobilization: 1,633; More Redemption: 1,157; Equal: 325

Statistical Analysis

Descriptive statistical techniques were used to understand the central tendency, variability, distribution shape, and extreme observations in the major numerical variables. The techniques include Mean, Median, Mode, Variance, Standard Deviation, Skewness, and Kurtosis.

Central Tendency

Variable	Mean	Median
net_aum	75,805.39	31,617.13
avg_net_aum	76,722.93	31,897.35
net_inflow_outflow	473.37	26.85
no_of_folios	2,518,408	521,606
fund_mobilized	20,263.59	1,022.55
repurchase_redemption	19,790.22	880.05

Most numerical variables show a considerable difference between their mean and median, indicating the presence of extreme high-value observations. The distributions are generally right-skewed, so the median can sometimes provide a more representative view of a typical observation than the mean.

Variance and Standard Deviation

Variable	Variance	Standard Deviation
no_of_folios	1.52×10^{13}	≈ 3.90 million
net_aum	$\approx 9.58 \times 10^{9}$	≈ 97,880
avg_net_aum	$\approx 9.99 \times 10^{9}$	≈ 99,935
fund_mobilized	$\approx 1.18 \times 10^{10}$	≈ 108,655
repurchase_redemption	$\approx 1.18 \times 10^{10}$	≈ 108,423
net_inflow_outflow	8.33×10^{7}	≈ 9,126

The number of folios has the highest variance, showing very uneven investor participation. Net AUM and Average Net AUM also show considerable variation. Fund mobilization and repurchase/redemption have similar levels of variability, while net inflow/outflow has comparatively lower variance. The number of folios also has the largest absolute spread in standard deviation.

Skewness

Variable	Skewness
net_aum	2.39
avg_net_aum	2.43
net_inflow_outflow	-3.42
no_of_folios	1.98
fund_mobilized	11.40
repurchase_redemption	11.29

Net AUM and Average Net AUM are positively skewed because a few schemes have very large AUM. Number of folios is positively skewed because some schemes have exceptionally high investor participation. Fund mobilization and repurchase/redemption are extremely positively skewed because a small number of observations have very large values. Net inflow/outflow has strong negative skewness because extreme negative outflows extend the distribution toward the left.

Kurtosis

Variable	Kurtosis
net_aum	7.73
avg_net_aum	7.75
net_inflow_outflow	108.32
no_of_folios	3.42
fund_mobilized	175.82
repurchase_redemption	172.09

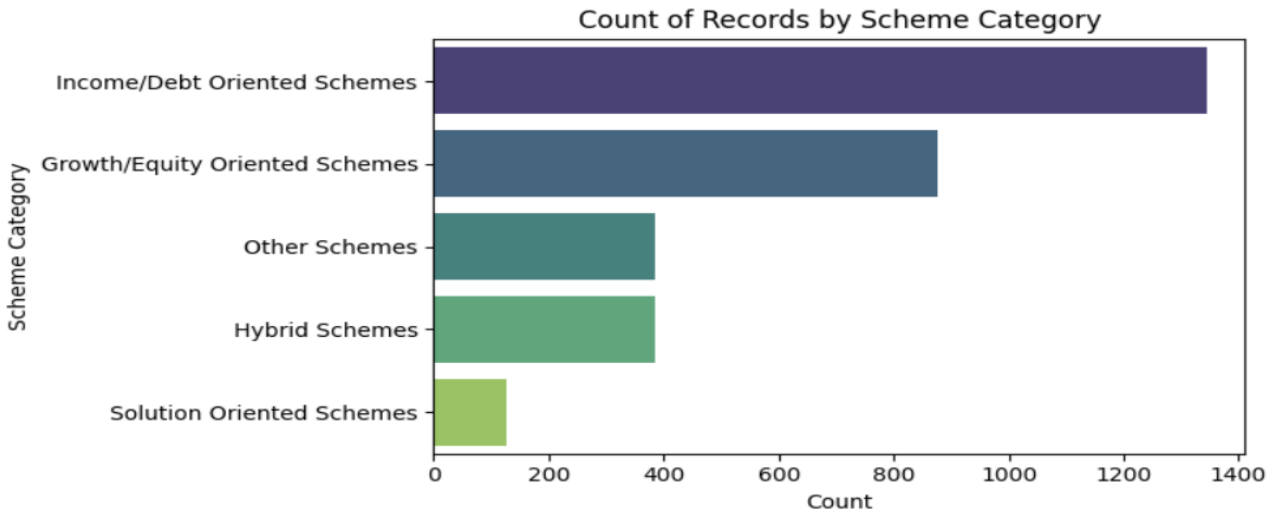
Net AUM and Average Net AUM have high kurtosis, indicating heavy tails. Net inflow/outflow has extremely high kurtosis, showing that most observations are concentrated around zero while a few extreme observations exist. Fund mobilization and repurchase/redemption also have very high kurtosis because occasional large transactions create extreme values. Number of folios has comparatively moderate kurtosis.

Exploratory Data Analysis (EDA)

The EDA is organized into univariate, bivariate, and multivariate analysis. The purpose is to understand distributions, category differences, trends, relationships, and combined effects of multiple variables.

Univariate Analysis — Scheme Category Distribution

Chart: Count of Records by Scheme Category

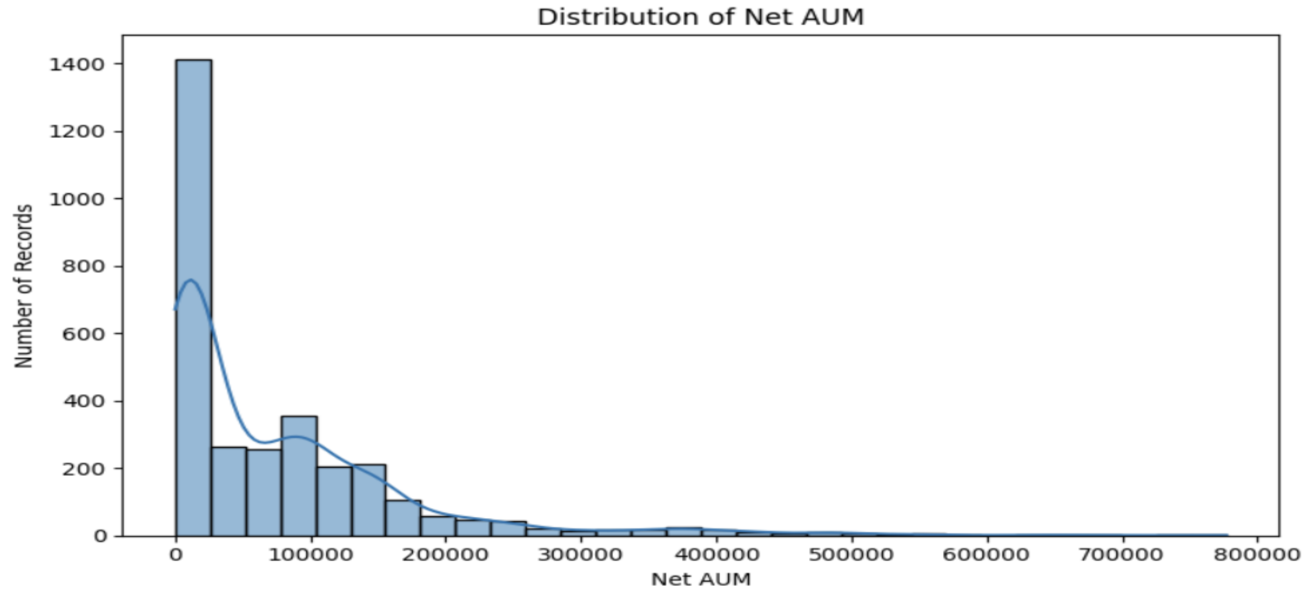


Income/Debt schemes dominate the dataset with approximately 1,400 records, followed by Equity schemes with approximately 900. Hybrid (approximately 400), Other (approximately 500), and Solution Oriented (approximately 150) are smaller groups.

Business insight: Debt schemes provide liquidity but are highly cyclical; Equity schemes show strong retail participation and drive asset growth; Hybrid schemes provide a balance of risk and return; Solution Oriented schemes are underutilized and require targeted promotion.

Univariate Analysis — Distribution of Net AUM

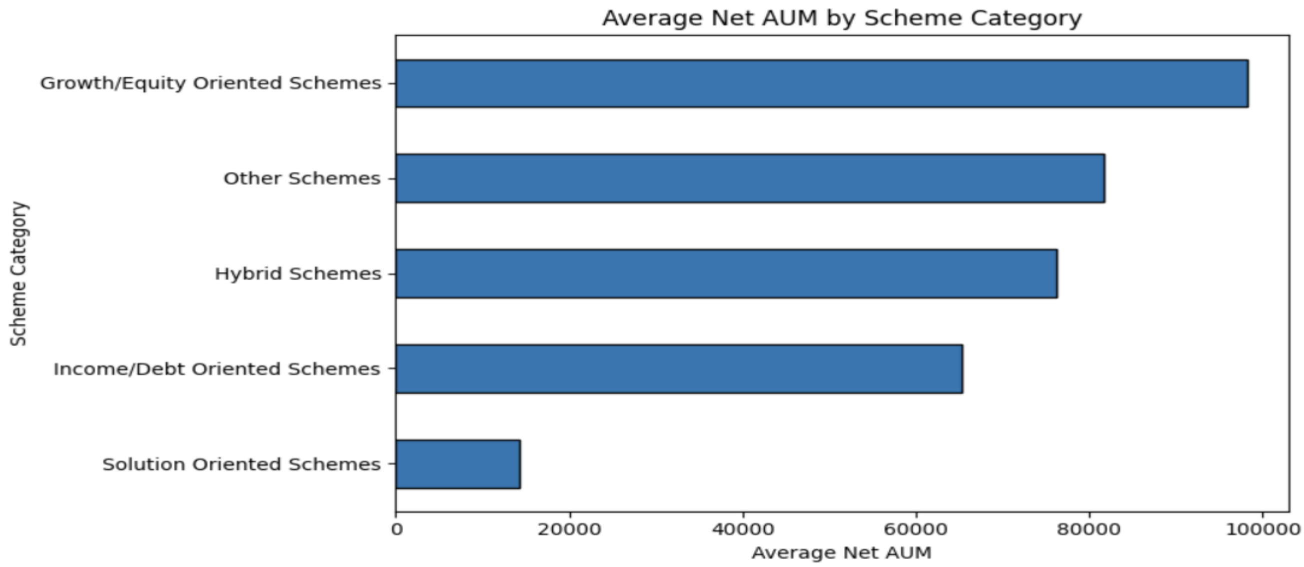
Chart: Distribution of Net AUM



Most schemes cluster at low Net AUM values between 0 and 50K. A smaller number of schemes extend into very high ranges of approximately 777K, creating a long right tail.

Business insight: Asset concentration is unequal, with a handful of large schemes dominating. Smaller schemes are common but contribute less to overall AUM. The analysis suggests focusing on scaling mid-tier schemes to reduce dependence on a few very large schemes.

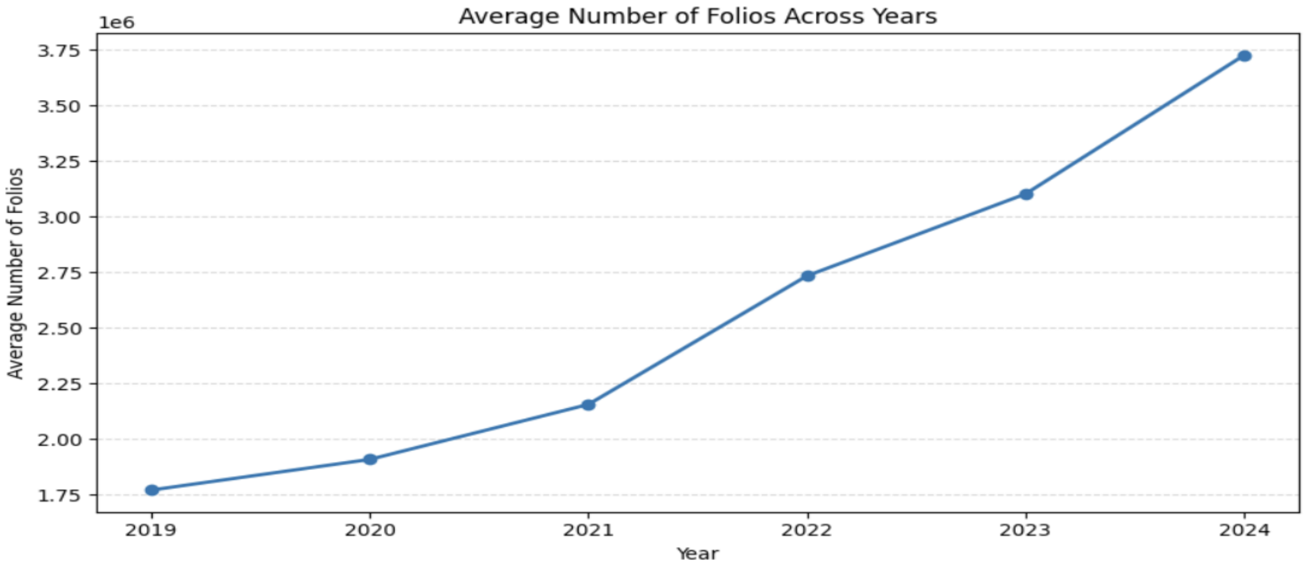
Bivariate Analysis — Average Net AUM by Scheme Category



Equity schemes lead at approximately 180K. Other schemes show a late surge to approximately 162K, Hybrid schemes reach approximately 126K, Debt schemes remain around 60–70K, and Solution Oriented schemes remain lowest at approximately 23K.

Business insight: Equity remains the growth engine. Other schemes present opportunities after 2021. Hybrid schemes act as reliable diversification products, while Debt schemes are stable but show less long-term growth. Solution Oriented schemes may require redesign or niche targeting.

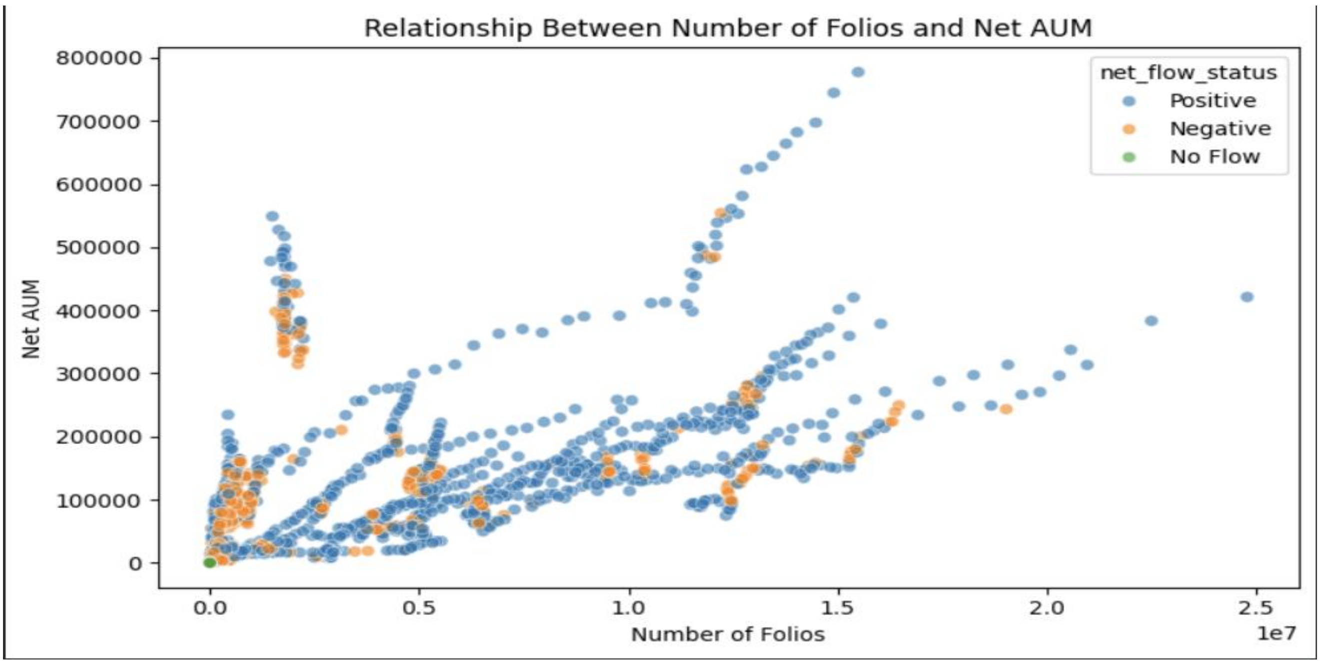
Bivariate Analysis — Average Number of Folios Across Years



Average folios rose steadily from approximately 1.8 million in 2019 to approximately 3.6 million in 2024. This indicates nearly doubled investor participation over the analyzed period.

Business insight: Continued digital onboarding and investor awareness initiatives can help sustain participation growth.

Bivariate Analysis — Folios and Net AUM



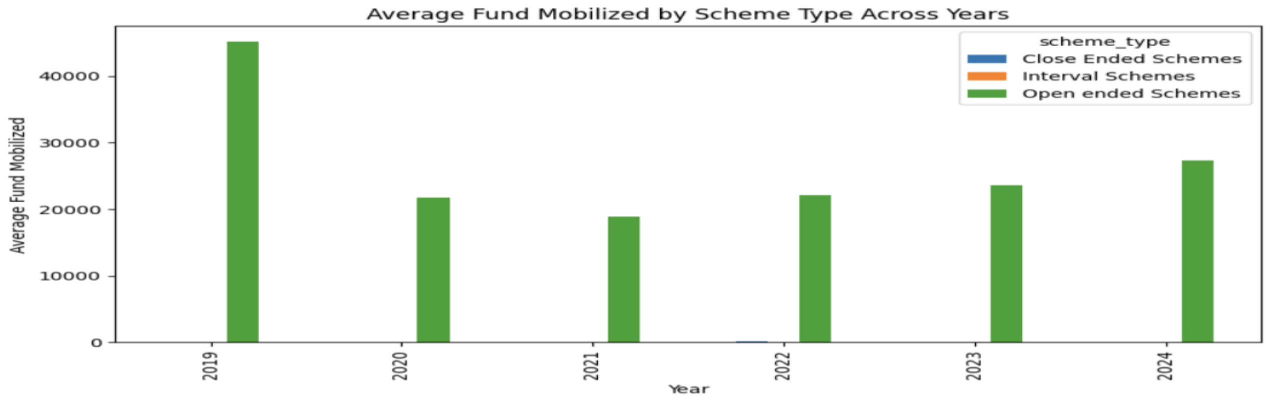
The analysis shows a positive relationship: higher numbers of folios generally correspond to higher Net AUM. At the same time, some schemes achieve large AUM with fewer folios, which is consistent with the influence of institutional investors.

Business insight: Retail participation contributes to overall growth, while institutional investors can skew aggregate results. Both segments should therefore be considered when interpreting AUM and folio relationships.

Multivariate Analysis

Multivariate analysis evaluates multiple variables together to understand how scheme type, category, year, investment activity, and investor participation interact.

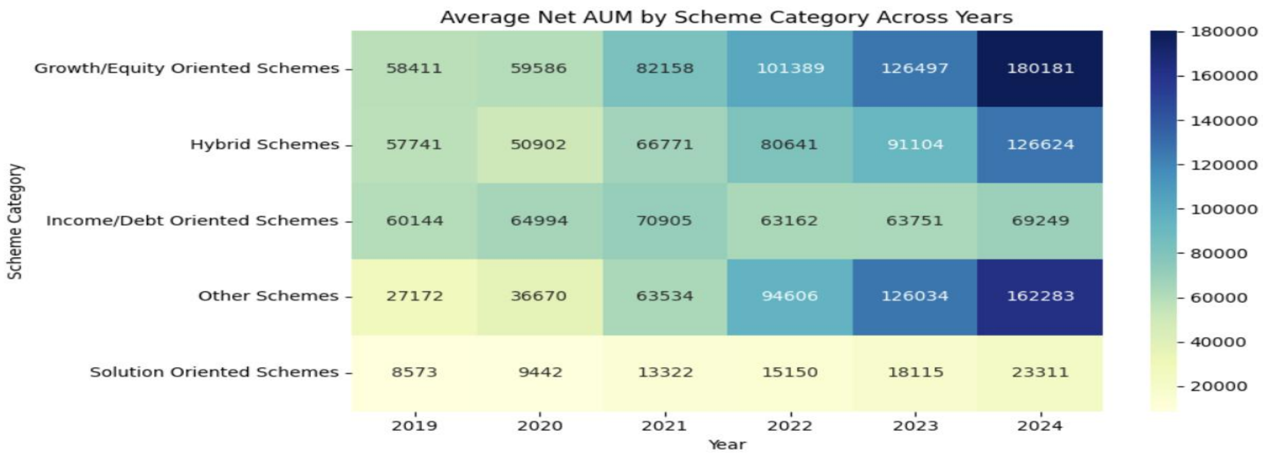
Average Fund Mobilized by Scheme Type Across Years



Open Ended schemes dominate fund mobilization in every year, reaching an approximate peak of 20K in 2019. Close Ended and Interval schemes are comparatively negligible.

Business insight: Open Ended schemes are the backbone of investment activity, while Close Ended and Interval schemes may require innovation or repositioning.

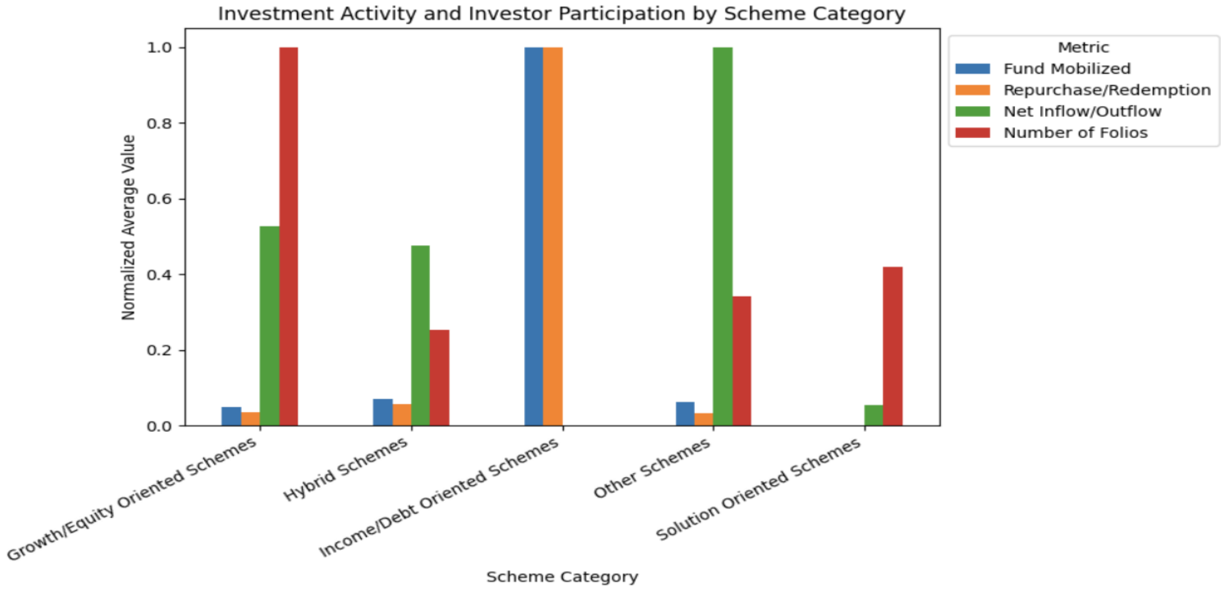
Average Net AUM by Scheme Category Across Years



Equity schemes increased from approximately 58K in 2019 to approximately 180K in 2024. Hybrid schemes rose steadily to approximately 126K. Debt schemes remained relatively stable at approximately 60–70K. Other schemes surged after 2021 to approximately 162K, while Solution Oriented schemes remained weak at approximately 23K.

Business insight: Equity and Hybrid categories consistently attract investors. Debt schemes provide stability, while Other schemes offer diversification opportunities and may be promoted further.

Investment Activity and Investor Participation by Scheme Category



The analysis compares normalized averages of Fund Mobilized, Repurchase/Redemption, Net Inflow/Outflow, and Number of Folios. Debt schemes show the highest mobilization and redemption, Equity schemes have the highest folios, Other schemes show the strongest net inflows after 2021, Hybrid schemes remain balanced across metrics, and Solution Oriented schemes have the lowest overall impact.

Business insight: Debt schemes drive liquidity cycles, Equity schemes dominate retail adoption, Other schemes show momentum, Hybrids provide balance and risk mitigation, and Solution Oriented schemes require strategic repositioning.

Key Findings

- **Mutual Fund Categories**: Income/Debt Oriented Schemes dominate the dataset in terms of record representation.
- **Asset Management**: Growth/Equity Oriented Schemes have the highest average Net AUM.
- **Investor Participation**: Investor participation increased steadily between 2019 and 2024.
- **Fund Mobilization**: Open Ended Schemes consistently dominate fund mobilization.
- **Fund Flows**: Other Schemes show strong net inflows, while Income/Debt Schemes have high mobilization and redemption activity.
- **AUM Growth**: Growth/Equity and Hybrid Schemes show strong growth in average Net AUM across the analyzed period.
- **Data Distribution**: Several financial variables are strongly skewed and contain extreme observations , making median and distribution-based analysis important.

Business Insights

The analysis suggests that different mutual fund categories play different roles in the mutual fund ecosystem.

Growth/Equity Schemes demonstrate strong asset growth and broad investor participation, while **Income/Debt Schemes** contribute significantly to fund mobilization and redemption activity.

The increase in folios from 2019 to 2024 indicates growing investor participation. The positive relationship between folios and Net AUM further suggests that increasing investor participation is generally associated with higher assets under management.

The high skewness and kurtosis in financial variables also show that a relatively small number of observations can have a major influence on overall averages. Therefore, business decisions should consider **median values, category-level comparisons, and distribution patterns** rather than relying only on mean values.

Mutual Fund Investment & Performance Analysis (2019–2024) Using Power BI

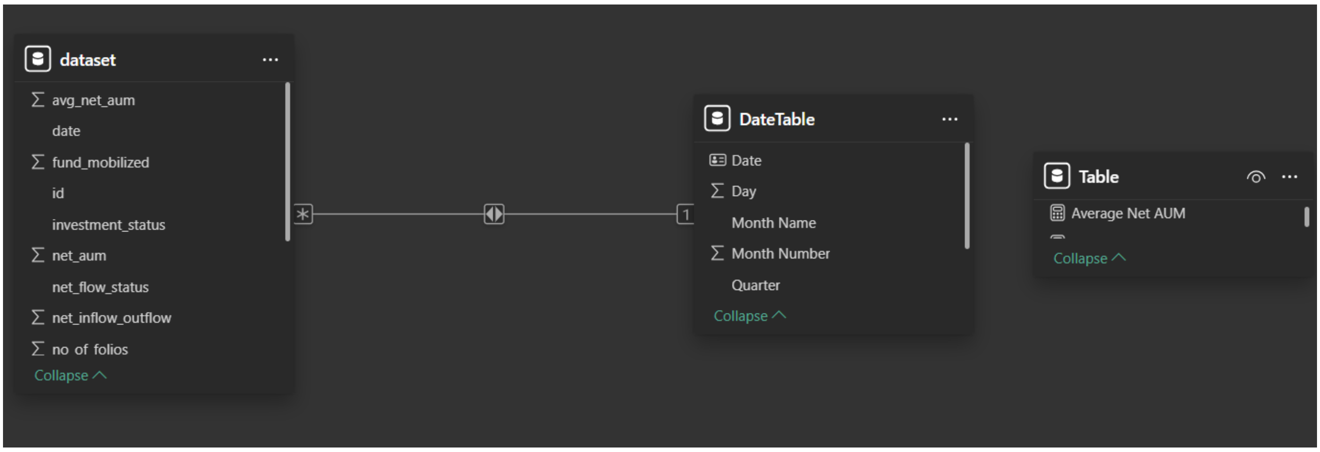
A Power BI version of the analysis was developed to present the results in an interactive, business-friendly format. The dashboard is designed to align with the Python analysis while emphasizing quick decision-making and visual storytelling.

Dashboard Objectives

- Provide executive-friendly KPIs for quick decision-making.
- Visualize scheme category distribution and dominance.
- Track Net AUM growth across years and categories.
- Compare fund mobilization versus redemption across scheme types.
- Explore investor participation trends through folio growth.
- Study relationships between folios, AUM, and fund flows.

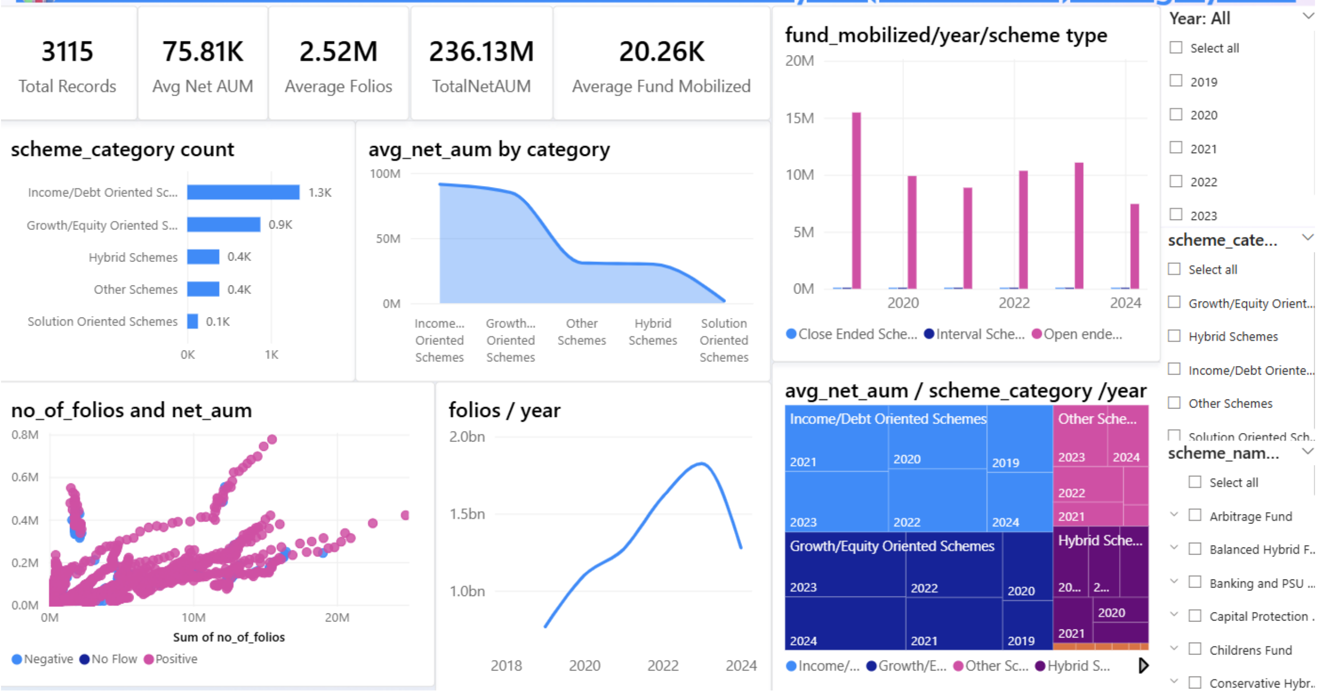
Data Modeling

The Power BI model uses a fact table containing the mutual fund dataset and a DateTable containing 2019–2024 date attributes such as Year, Month, Quarter, and Weekday. The relationship is dataset[date] → DateTable[Date] with a one-to-many relationship.



Dashboard

📊 Mutual Fund Investment & Performance Analysis (2019–2024) Using Python



Dashboard Features

KPI Cards → Total Records, Avg Net AUM, Total Net AUM, Average Folios, Average Fund Mobilized.

Category Analysis → Bar chart of scheme counts, treemap of Net AUM by category & year.

Trend Analysis → Line chart of folios growth (2019–2024), Net AUM trends.

Fund Flow Analysis → Bar chart of mobilization vs redemption by scheme type.

Correlation Analysis → Scatter plot of folios vs Net AUM, colored by flow status.

Power BI Insights

- Equity schemes dominate Net AUM growth, followed by Debt and Hybrid.
- Open Ended schemes drive most mobilization, while Close Ended and Interval schemes remain minor.
- Investor folios doubled between 2019 and 2024, showing strong participation.
- Net inflow/outflow patterns reveal liquidity cycles, with sharp outflows in certain years.
- Higher folios generally correspond to higher AUM, although institutional schemes can achieve large AUM with fewer folios.

Tools, Technologies and Analysis Techniques

Area	Tools / Techniques
Programming	Python 3.10+
Data Manipulation	Pandas
Numerical Analysis	NumPy
Visualization	Matplotlib, Seaborn
Python Environment	Google Colab

Business Intelligence	Power BI Desktop
Dashboard Calculations	DAX

Analysis techniques used include Data Inspection, Data Cleaning, Missing Value Analysis, Duplicate Detection, Data Validation, Feature Engineering, Descriptive Statistics, Central Tendency, Variance, Standard Deviation, Skewness, Kurtosis, Univariate Analysis, Bivariate Analysis, Multivariate Analysis, Trend Analysis, Correlation/Relationship Analysis, and Business Insight Generation.

Outcome

The Power BI dashboard provides a clear, interactive view of mutual fund performance, aligning with the Python analysis but tailored for business storytelling. It enables mentors and stakeholders to quickly grasp totals, trends, and category dominance.

Conclusion

This project demonstrates how Python can be used to analyze real-world mutual fund data and transform raw financial records into meaningful insights. The analysis identifies important patterns in Net AUM, fund mobilization, redemption, investor participation, and net fund flows across different scheme categories and reporting years. Overall, the project highlights the dominance of Open Ended Schemes in fund mobilization, the strong asset growth of Growth/Equity Schemes, increasing investor participation, and significant variability in financial metrics. The combination of statistical analysis, EDA, visualization, and Power BI dashboarding provides a structured approach to understanding mutual fund investment behavior and presenting findings in a business-oriented manner.